

PROGRAM A: FIFO PAGE REPLACEMENT

C PROGRAM

```
#include <stdio.h>
```

```
int main() {  
    int pages[50], frames[10];  
  
    int n, f, i, j, k = 0;  
  
    int fault = 0, found;  
  
    printf("Enter Number of Pages: ");  
    scanf("%d", &n);  
  
    printf("Enter Reference String:\n");  
    for (i = 0; i < n; i++) {  
        scanf("%d", &pages[i]);  
    }  
  
    printf("Enter Number of Frames: ");  
    scanf("%d", &f);  
  
    for (i = 0; i < f; i++) {  
        frames[i] = -1;  
    }  
  
    for (i = 0; i < n; i++) {  
        found = 0;  
        for (j = 0; j < f; j++) {  
            if (frames[j] == pages[i]) {  
                found = 1;  
                break;  
            }  
        }  
  
        if (found == 0) {  
            frames[k] = pages[i];
```

```

        k = (k + 1) % f;

        fault++;

    }

}

printf("Total Page Faults = %d\n", fault);

return 0;

}

```

SHELL SCRIPT

```

GNU nano 8.7.1
#!/bin/bash

echo "FIFO Page Replacement Demonstration"

pages=(7 0 1 2 0 3 0 4 2 3 0 3 2)
frames=3

echo "Reference String: ${pages[@]}"
echo "Frames: $frames"

echo "FIFO Algorithm Executed"

```

Output :

```

(ganesh@LAPTOP-R3RDH9NG)~$ nano fifo.sh

(ganesh@LAPTOP-R3RDH9NG)~$ chmod +x fifo.sh

(ganesh@LAPTOP-R3RDH9NG)~$ bash fifo.sh
FIFO Page Replacement Demonstration
Reference String: 7 0 1 2 0 3 0 4 2 3 0 3 2
Frames: 3
FIFO Algorithm Executed

```